

Alvin Reji

Boston, MA | (516) 424 0863 | reji.alv@northeastern.edu | [linkedin.com/in/alvin-reji](https://www.linkedin.com/in/alvin-reji)

EDUCATION

Northeastern University, Boston, MA

Sep 2024 – Dec 2027

Candidate for Bachelor of Science in Mechanical Engineering with an Aerospace Minor

GPA: 3.87

Activities: AerospaceNU, ASME, Materials Engineering Research, TAMID, InterVarsity

Relevant Courses: Statics, Mechanics of Materials, Dynamics, Propulsion, Mechanical Design, Thermodynamics, Fluid Mechanics

University at Buffalo, Buffalo, NY

Aug 2023 – May 2024

Activities: University Nanosatellite Program, Solar Splash Electric Boat Racing, ASME

GPA: 4.0

SKILLS

Hardware: Microcontrollers, CNC Routing, 3D Printing, TIG Welding, Circuit Design, Soldering, Laser Cutting, Oscilloscope, ShopBot

Software: SolidWorks, Ansys, Minitab, MS Office, GSuite, Altium, MATLAB, AutoCAD, Python, C++, MATLAB, CAD, FEA, CFD

Interests: Rock climbing, Weightlifting, Volleyball, Spikeball, Cycling, Running, Hiking, Billiards, Photography, Videography, Pottery

EXPERIENCE

Johnson & Johnson Medtech

Danvers, MA

Product Development Engineering Co-Op

Jan 2026 – Present

- Designed and iterated custom SolidWorks test fixtures to support cardiac pump mechanical testing for DV and TMV protocols
- Evaluated device safety and usability via Instron testing and Minitab statistical analysis for data-driven design improvements
- Built a Python/OpenCV computer vision pipeline to extract experimental metrics and inform cross-functional design decisions

Massachusetts Institute of Technology

Cambridge, MA

Teaching Assistant

Aug 2025 – Present

- Educated 24 students on the iterative design process while introducing Six Sigma quality concepts to improve process reliability
- Fostered a collaborative, inclusive classroom by coaching conflict resolution, and promoting effective decision-making strategies

AerospaceNU CubeSAT at Northeastern University

Boston, MA

Mechanical Engineer

Sep 2024 – Dec 2025

- Designed optical component casings for 1U CubeSAT, ensuring structural integrity under launch and orbital conditions using FEA
- Generated ASME Y14.5-compliant engineering drawings using GD&T principles for high tolerance fabrication and assembly
- Fabricated prototypes using FDM and SLA 3D printing, conducting F3 verification and risk analysis prior to final machining

AerospaceNU Rocket Team at Northeastern University

Boston, MA

Avionics Engineer

Sep 2024 – Dec 2025

- Developed software for reliable sensor data acquisition on flight computers by implementing signal processing and noise reduction
- Designed flight computer PCBs optimized for power distribution, EMI compliance, and thermal management using KiCAD

Material Engineering Research at Northeastern University

Boston, MA

Undergraduate Research Assistant

Sep 2024 – Present

- Operated volumetric 3D printers to fabricate hierarchical microstructures with micron-level precision for fine-feature applications
- Engineered 30+ experimental samples using photopolymer resins and polymers to characterize mechanical/material properties

Solar Splash Electric Boat Racing at University at Buffalo

Buffalo, NY

Electrical Engineering Subsystem Lead

Aug 2023 – May 2024

- Led 7 engineers in designing and integrating an electric boat power system, achieving full NESC safety standard compliance
- Directed testing and troubleshooting, while mentoring team members on best practices to ensure reliable power system operation
- Optimized power system architecture, reducing overall cost by \$7,000 through component selection reviews and design analyses

University Nanosatellite Program at University at Buffalo

Buffalo, NY

Guidance, Navigation, & Control Engineer

Aug 2023 – May 2024

- Developed MATLAB simulations for satellite missions, modeling orbital mechanics, attitude dynamics, and control responses
- Implemented Monte Carlo algorithms and Kalman filters to improve satellite state determination and reduce noise by 54%
- Authored interface control documents and led hardware trade studies on GN&C component viability for system integration

PROJECTS

GE1501 Cornerstone of Engineering at Northeastern University

Boston, MA

Horse Racing Tracking Collar

Oct 2024 – Dec 2024

- Designed and fabricated an ESP32-based collar using SolidWorks and FDM 3D printing, ensuring sensor stability and durability
- Implemented GPS and accelerometer integration, enabling accurate horse position and motion tracking with ± 1 m positional error
- Developed embedded C++ software for real-time Wi-Fi data transmission and data visualization through a custom PyQt5 GUI

NASA Techrise Student Challenge

Elmont, NY

Atmospheric Measurement of Noxious Gases Using Sensors (AMONGUS)

Nov 2021 – July 2022

- Conducted pre-launch testing in low-pressure chambers, improving payload reliability by 30% under high altitude conditions
- Fabricated and assembled payload components using CNC machining and laser cutting, ensuring structural integrity during flight